

Inspect the workbook before changing any data. Explain what each dataset represents, its likely unit of observation and how the datasets relate

1.1) Learners Dataset

This dataset represents the learners' details. The unit of observation is an individual learner.

The fields include:

- Learner ID [**learner_id**]
- Gender [**gender**]
- Grade [**grade**]
- Registration Date [**registration_date**]

1.2) Engagement Dataset

This dataset represents how much each student engages with each of their academic subjects. The unit of observation is an individual learner, along with their set of subjects and engagement metrics for each subject.

The fields include:

- Learner ID [**learner_id**]
- Subject [**subject**]
- Tutoring Attendance Percentage [**tutoring_attendance_pct**]
- Number of Videos Watched [**videos_watched**]
- Watch Time (in minutes) [**watch_time_minutes**]

1.3) Assessment Dataset

This dataset represents how each student scored on their assessments for different subjects. The unit of observation is an individual learner, along with their set of subjects and assessment marks for each subject.

The fields include:

- Learner ID [**learner_id**]
- Subject [**subject**]
- Baseline Assessment Mark [**baseline_mark**]

- Latest Assessment Mark [**latest_mark**]

1.4) Support Capacity

This dataset contains tutoring data for each academic subject. The unit of observation is an academic subject.

The fields include:

- Subject [**subject**]
- Active Tutors [**active_tutors**]
- Weekly Tutoring Hours [**weekly_tutor_hours**]
- Waiting List (I'm assuming that's the number of students in that waiting list) [**waiting_list**]
- Estimated Cost Per Extra Hour [**estimated_cost_per_extra_hour**]

1.5) Enquiries

I'm assuming that this dataset contains a record of enquiries where current or new students ask about a specific subject and its requirements, such as pass marks from previous grades, expectations once enrolled, etc. The unit of observation is an individual enquiry.

The fields include:

- Enquiry [**enquiry_id**]
- Enquiry Date [**enquiry_date**]
- Grade [**grade**]
- Subject [**subject**]
- Enquiry Converted to Registration (Meaning that the student was interested into enrolling or registering to learning the subject) [**converted_to_registration**]

1.6) How each datasets relate

1. Learner ID link
 - a. The learners, engagement, and assessment datasets are explicitly linked by the learner ID field
2. Assessment, support, and enquiries link
 - a. The subject field in the 'support_capacity' dataset is linked to the student's assessment performance in the 'assessments' dataset, the engagements of students in the 'engagement' dataset, as well as the developing interests of students in a specific subject in the 'enquiries' dataset.

Perform a data-quality audit. Identify and quantify important quality problems and explain how they could distort analysis or the final business recommendation.

2.1) Learner's dataset

1. Gender field
 - Inconsistent formatting in most entries
 - Contains missing values in a few entries
2. Learner ID field
 - Inconsistent formatting found in one entry
3. Grade field
 - Inconsistent formatting
 - Incorrect data types used in some entries
4. Registration date field
 - Inconsistent date-time formatting in some entries
 - Contains missing values in some entries

Impacts:

- **Gender:** Inconsistent formatting and missing values prevent accurate demographic segmentation, which can obscure critical gender-based performance trends in your final analysis.
- **Learner ID:** Formatting inconsistencies in this primary key hinders reliable table joins, potentially creating orphaned records and an incomplete view of the learner's journey.
- **Grade:** Incorrect data types (object instead of integer) and inconsistent formats make it impossible to mathematically aggregate metrics or accurately group learners by their academic level.
- **Registration date:** Mixed date-time formats and missing data disrupt time-series analysis and cohort tracking, complicating the evaluation of seasonal registration campaigns.

2.2) Engagement dataset

1. Subject field
 - Inconsistent formatting
 - Contains missing values in 2 entries
2. Tutoring attendance percentage field
 - Contains missing values in some entries

- Inconsistent data types in some entries. Most entries use floating point data types. Some entries use incorrect decimal point symbols. Some entries use percentage symbols.
 - Contains negative values in one entry
 - Multiple entries exceed the realistic limit of 100
3. Videos watched field
- Contains negative values in some entries
 - Object data types used in some entries instead of using integers
 - Contains missing values in one entry
4. Watch time minutes
- Contains negative values in multiple entries
 - Incorrect floating point (decimal) symbol (',' instead of '.') used in multiple entries

Impacts:

- **Subject:** Formatting inconsistencies and missing values fragment subject groupings, distorting the perceived popularity and engagement levels of specific topics.
- **Tutoring attendance percentage:** Missing data, negative values, mixed types, and impossible metrics (exceeding 100%) will drastically skew average attendance rates, leading to misguided business strategies regarding tutoring interventions.
- **Videos watched:** Negative values, incorrect object data types, and missing entries invalidate content consumption metrics, masking the true effectiveness of the platform's video materials.
- **Watch time minutes:** Negative entries and incorrect decimal symbols corrupt total watch time aggregations, leading to inaccurate conclusions about overall learner dedication.

2.3) Assessments dataset

1. Learner ID field
 - Contains missing values
 - Inconsistent formatting in two entries
2. Subject field
 - Inconsistent formatting in multiple entries
3. Baseline mark field
 - Negative value in one entry
 - Inconsistent formatting in multiple entries (percentage symbol used instead of integer data type)
 - Contains missing value in multiple entries
4. Latest mark field
 - Multiple entries exceed the limit of 100

- Three entries are using object data type instead of integer data type

Impacts:

- **Learner ID:** Missing values and formatting inconsistencies break the crucial link between learners and their scores within the workbook, rendering those specific records useless for individual progress tracking.
- **Subject:** Inconsistent formatting splinters categories, unnecessarily complicating and distorting cross-subject performance comparisons.
- **Baseline mark:** Negative values, missing data, and incorrect percentage formatting corrupt the starting benchmark for learners, making accurate progress measurement impossible.
- **Latest mark:** Values exceeding 100 and incorrect object data types artificially inflate academic performance averages and will break automated calculation pipelines.

2.4) Support capacity dataset (All records have duplicates)

1. Subject field
 - Inconsistent formatting in multiple entries
2. Weekly tutor hours field
 - Contains negative value in one entry
 - One entry uses and incorrect data type (object) instead of integer
3. Waiting list field
 - Contains missing value in one entry
4. Estimated cost per extra hour field
 - Inconsistent data types used (Object and floating point) instead of integer in two entries

Impacts:

- **General Duplicates:** Having duplicates across all records fundamentally doubles the perceived capacity, causing the business to severely overestimate available resources and over-commit to learners.
- **Subject:** Inconsistent formatting misaligns capacity metrics with actual subject demand, making it difficult to assign tutors efficiently.
- **Weekly tutor hours:** Negative values and incorrect object types skew the calculation of available tutoring time, posing a high risk of understaffing.
- **Waiting list:** Missing values obscure the true backlog of learner demand, preventing timely responses to tutoring bottlenecks.
- **Estimated cost per extra hour:** Mixed data types complicate budget forecasting and financial modeling for scaling additional support.

2.5) Enquiries dataset

- Enquiry date field
 - Incorrect date-time format in one entry
- Grade field
 - Incorrect data type used (object) instead of integer in one entry
- Subject field
 - Inconsistent formatting found in one entry
- Converted to registration field
 - Inconsistent formatting in multiple field (1 should be converted to yes, and 0 should be converted to no)

Impacts:

- **Enquiry date:** Incorrect date-time formats disrupt lead-time analysis, making it harder to track the speed and efficiency of the conversion funnel.
- **Grade:** Using an object data type prevents standard numeric sorting, grouping, or filtering of prospective learners.
- **Subject:** Inconsistent formatting misrepresents the true volume of interest in specific subjects, potentially skewing the focus.
- **Converted to registration:** Mixing binary logic (1/0) with text (yes/no) complicates the accurate calculation of conversion rates.

Cleaning and standardizing the data

3.1) Learners dataset

Learner ID Column

- **Problem:**
 - Inconsistent formatting (e.g., using "lnd" instead of "LND").
 - Missing values.
- **Action:**
 - Apply an uppercase function to rows where the Learner ID begins with "lnd".
 - Remove "LND9999"
 - Drop those missing values since,
 - 1) There were only 2 rows with missing values out of 146. Which means most of the data integrity and accuracy will remain unaffected.

- 2) We cannot simply impute it “LND” plus random numbers as that would cause some errors when we finally must join this dataset with other datasets using learner ID as the primary key.
- **Justification:**
 - It is important that every Learner ID follows the same structural format so it can serve as a primary key. This will enable us to link this dataset to other related datasets, such as the "Engagement" and "Assessments" datasets. Records for “LND9999” contain little to no information/data to work with and will not have a noticeable effect when dropped.

Gender Column

- **Problem:**
 - Inconsistent data entry.
 - Missing values.
- **Action:**
 - Replace irregular entries (e.g., "f", "MALE") with standardized terms ("Male" and "Female").
 - Replace "Unknown" with "NaN".
 - Replace missing values with "Prefer not to say"
- **Justification:**
 - Standardizing this column to "Male", "Female", "Prefer not to say", and "NaN" ensures consistency during the data analytics process. Failing to do so could lead to inaccurate insights caused by minor typos.

Grade Column

- **Problem:**
 - Data inconsistencies.
 - The column is formatted as an object data type instead of an integer.
- **Action:**
 - Standardize text-based entries (e.g., "Grade 11", "G11", "Grade Eleven") to their corresponding integer values (e.g., 11).
 - Convert the data type of the entire column from an object to an integer.
- **Justification:**
 - This ensures an accurate representation of the different grade populations and prevents formatting errors from creating artificial outliers (e.g., recording a "Grade 13" when the realistic limit is Grade 12).

Registration Date Column

Problem:

- Inconsistent datetime formats (the correct format is YYYY-MM-DD).
- Incorrect placement of "NaN" or null values.
- The column is formatted as an object instead of a datetime64 data type.
- **Action:**
 - Find incorrectly formatted dates and convert them to the standard YYYY-MM-DD format.
 - Replace text strings like "not_a_date" with "NaN".
- **Justification:**
 - Inconsistent date formatting can cause inaccurate timeline representations during analysis, leading to misleading insights. Additionally, some data analytics software may not natively support automatic date conversion; standardizing the format reduces manual intervention and saves time for actual analysis.

3.2) Engagement dataset

Learner ID Column

- Problem:
 - Duplicate rows
 - Irrelevant/ incorrect learner ID structure ('TEMP-004', 'LND9999')
 - Missing values
- Action:
 - Drop entire rows if their learner ID and other columns have the same values
 - Drop rows whose learner ID equates to "TEMP-004", "LND9999", or has missing values
- Justification:
 - Not dropping duplicate rows can inflate certain metrics and cause misleading insights. LND9999 and TEMP-004 seem like they are data entry errors as one does not follow the correct identification structure, while LND9999 does not have a lot of data within its corresponding fields. Rendering them unusable.

Subject Column

- Problem:
 - No consistent format

- It is unclear where “Sci” either belongs to “Life Science” or “Physical Science”
- Action:
 - Initially, I planned to manually impute “Sci” subject values as either "Life Sciences" or "Physical Sciences" using the ***tutoring_attendance_pct***, ***videos_watched***, and ***watch_time_minutes*** columns. For rows with complete data, if the ***videos_watched*** mean value for Life Sciences was higher than that of Physical Sciences, I would compare the labeled "Sci" row against those averages. If its value was closer to the Life Science average, I would classify that row as Life Science. The same logic would apply across the other features.
 - However, doing this manually would be time-consuming and tedious. Instead, I decided to use a machine learning classification algorithm to handle the task, specifically, the **Random Forest Classifier** from Scikit-learn.
 - I trained the model to classify whether an entry corresponds to Life Sciences or Physical Sciences based on ***tutoring_attendance_pct***, ***videos_watched***, and ***watch_time_minutes***. Once trained, I filtered out the generic "Sci" rows and passed their feature data through the model to predict the correct subject.
 - But before I could train the model, I first must clean the other columns.
- Justification:
 - Let’s be honest, you’d rather come up with a fun and creative way of doing this entire process of imputation than having to do it safely and manually.

Column name: **tutoring_attendance_pct**

- Problem:
 - Percentage symbols were included in some values.
 - Negative values were present.
 - Some values exceeded the valid maximum of 100%.
- Action:
 - Removed the % symbol.
 - Converted the column to a numeric data type.
 - Converted negative values to positive values using the absolute value.
 - Replaced values above 100 with 100.
- Justification:
 - Attendance percentages must fall between 0 and 100. Removing formatting characters allows numerical analysis, while correcting negative and excessive values ensures that the measurements remain within a valid percentage range.

Column name: videos_watched

- Problem:
 - Missing values were present.
 - Negative values were present.
 - The column was stored as a non-integer numeric type.
- Action:
 - Checked the column for missing values.
 - Converted negative values to positive values using the absolute value.
 - Converted the values to floating-point numbers.
- Justification:
 - The number of videos watched cannot be negative. Converting the values to numeric form makes the column suitable for calculations and machine-learning analysis. Missing values were retained so they could be handled by the model's median imputation step.

Column name: watch_time_minutes

- Problem:
 - Decimal values used commas instead of periods.
 - Negative values were present.
 - The column required numeric conversion.
 - Missing values were present.
- Action:
 - Replaced commas with periods.
 - Converted the values to floating-point numbers.
 - Converted negative values to positive values using the absolute value.
 - Checked the descriptive statistics and number of null values.
- Justification:
 - Watch time must be numeric and non-negative. Standardizing decimal notation makes the values interpretable by Python and machine-learning algorithms, while checking null values confirms the remaining data-quality issues before modelling.

3.3) Assessments dataset

Column name: learner_id

- Problem:

- Some learner IDs used lowercase letters.
- The placeholder ID LND9999 was present.
- Action:
 - Removed rows where learner_id was LND9999.
 - Converted all learner IDs to uppercase.
 - Checked for IDs that did not begin with LND.
- Justification:
 - Learner IDs are used to link records between datasets, so they must follow one consistent format. Removing the placeholder ID prevents invalid records from being included, while uppercase conversion ensures reliable matching.

Column name: subject

- Problem:
 - Subjects were represented using inconsistent abbreviations, spelling, capitalization, and plural forms.
 - Examples included ACC, Math, Maths, MATHEMATICS, Life Sciences, life sciences, Physical Sciences, and phys sci.
 - The value Sci was ambiguous.
- Action:
 - Standardized all identifiable values to Accounting, Mathematics, Life Science, and Physical Science.
 - Created a science-subject lookup from the cleaned engagement dataset using learner_id.
 - Used the learner ID to assign the corresponding science subject to assessment records labelled Sci.
- Justification:
 - Consistent subject labels are necessary for comparing assessment and engagement records. Using the learner ID to resolve Sci preserves the relationship between datasets and provides a more reliable classification than guessing from the assessment data alone.

Column name: latest_mark

- Problem:
 - Percentage symbols were included.
 - The value MISSING represented missing data.

- Negative marks were present.
- Some marks exceeded the valid maximum of 100.
- Values required numeric conversion.
- Action:
 - Removed percentage symbols.
 - Replaced MISSING with NaN then NaN with average mark.
 - Converted the column to nullable integer format.
 - Converted negative marks to positive values.
 - Replaced marks greater than 100 with 100.
- Justification:
 - Assessment marks should be numeric percentages between 0 and 100. These transformations preserve missing information while ensuring that the remaining values are valid and suitable for analysis.

Column name: baseline_mark

- Problem:
 - Percentage symbols were included.
 - The value MISSING represented missing data.
 - Negative marks were present.
 - Some marks exceeded 100.
 - Values required numeric conversion.
- Action:
 - Removed percentage symbols.
 - Replaced MISSING with NaN then NaN with the average baseline mark.
 - Converted the column to nullable integer format.
 - Converted negative values to positive values.
 - Replaced values above 100 with 100.
- Justification:
 - Standardizing baseline marks ensures that they can be compared consistently with latest marks. Restricting valid marks to the range 0 to 100 prevents invalid values from distorting performance analysis.

3.4) Support Capacity Dataset

Column name: subject

- Problem:
 - Subject names used plural forms instead of the standardized subject names.
 - Maths, Physical Sciences, and Life Sciences were inconsistent with the other datasets.
- Action:
 - Replaced Maths with Mathematics.
 - Replaced Physical Sciences with Physical Science.
 - Replaced Life Sciences with Life Science.
- Justification:
 - Using the same subject naming convention across all datasets makes it possible to join, compare, and aggregate support-capacity information reliably.

Column name: active_tutors

- Problem:
 - The column required a consistent numeric data type.
- Action:
 - Converted the column to nullable integer format.
- Justification:
 - The number of active tutors represents a whole-number count. Integer formatting prevents inconsistent data types while allowing missing values to remain represented as nulls.

Column name: weekly_tutor_hours

- Problem:
 - The column required a consistent numeric data type.
- Action:
 - Converted the column to nullable integer format.
- Justification:
 - Weekly tutor hours are recorded as whole-number quantities in the dataset. Converting the values to integers supports accurate calculations and consistent reporting.

Column name: waiting_list

- Problem:
 - The column required a consistent numeric data type.
- Action:
 - Converted the column to nullable integer format.
- Justification:
 - Waiting-list entries represent a count of learners, so integer formatting is appropriate and improves consistency for analysis.

Column name: estimated_cost_per_extra_hour

- Problem:
 - The column required a consistent numeric data type.
- Action:
 - Converted the column to nullable integer format.
- Justification:
 - The cost values were treated as whole monetary amounts. Standardizing the data type allows costs to be compared and used in calculations consistently.

Dataset-level issue: duplicate or incomplete rows

- Problem:
 - Each subject appeared to have duplicate rows.
 - Some rows contained missing values or more errors than the complete rows.
- Action:
 - Retained rows with indexes 0, 1, 2, and 3.
 - Removed rows with indexes 4, 5, 6, and 7.
 - Reset the index after removing the rows.
- Justification:
 - Keeping the complete, lower-error record for each subject prevents duplicate subjects from inflating the support-capacity results. Resetting the index maintains a clean sequential structure.

3.5) Enquiries dataset

Column name: enquiry_date

- Problem:
 - One date used the format 14/06/2026 instead of the standardized ISO format.
- Action:
 - Replaced 14/06/2026 with 2026-06-14.
- Justification:
 - Using the ISO date format makes dates consistent across the dataset and supports reliable sorting, filtering, and date-based analysis.

Column name: grade

- Problem:
 - Grade values included the text prefix Grade.
 - Some grade values were greater than the valid maximum of Grade 12.
 - The column required numeric conversion.
- Action:
 - Removed the Grade prefix.
 - Converted the values to nullable integers.
 - Replaced grades greater than 12 with 12.
 - Checked the descriptive statistics after cleaning.
- Justification:
 - Grades should be represented as numeric educational levels. Limiting values to Grade 12 prevents impossible grade values from affecting learner analysis.

Column name: subject

- Problem:
 - Subject names used inconsistent capitalization, abbreviations, and plural forms.
 - Examples included Maths, maths, phys sci, Physical Sciences, and Life Sciences.
- Action:
 - Standardized the values to Mathematics, Physical Science, and Life Science.
 - Checked the unique values after replacement.
- Justification:
 - Standardized subject labels ensure that enquiries can be compared with learner, engagement, assessment, and support-capacity records.

Column name: converted_to_registration

- Problem:
 - Boolean responses were represented using both letters and numbers.
 - Values included Y, N, 1, and 0.
- Action:
 - Replaced Y and 1 with Yes.
 - Replaced N and 0 with No.
 - Checked the unique values after standardization.
- Justification:
 - Converting all responses to the same Yes/No format removes ambiguity and makes registration-conversion rates easier to calculate and interpret.

Cleaned Data Validation Rules

Learner IDs

Rule:

- Must be non-null.
- Must follow the format LND followed by four digits, for example LND0001.
- IDs must be uppercase.
- Placeholder IDs such as LND9999 must not be present.

Validation:

- Learner IDs were converted to uppercase.
- Missing learner IDs were removed from the learners and engagement datasets.
- TEMP-004 and LND9999 were removed where identified.
- Remaining IDs should be checked against the pattern LNDxxxx.

Grades

Rule:

- Must be numeric integers.
- Valid values must be between 1 and 12.
- Missing values may be retained if no reliable replacement exists.

Validation:

- Grade labels were converted to integers.
- Enquiry grades above 12 were capped at 12.
- The cleaned grades should contain no values below 1 or above 12.

Subjects

Rule:

Subjects must belong to:

- Accounting
- Life Science
- Mathematics
- Physical Science

Validation:

- Abbreviations, spelling variations, plural forms, and capitalization differences were standardized.
- Ambiguous Sci values in the engagement data were classified using engagement features.
- Ambiguous Sci values in the assessment data were aligned using learner_id.
- All remaining subject values should belong to the approved list.

Dates

Rule:

- Dates must be valid calendar dates.
- Dates should use the YYYY-MM-DD format.
- Missing dates may be retained as NaN.

Validation:

- Irregular date formats were converted to ISO format.
- Invalid text such as not_a_date was replaced with NaN.
- The remaining non-null dates should be valid dates in 2026.

Assessment Marks

Rule:

- Values must be numeric.
- Valid marks must be between 0 and 100.
- Missing marks are represented as NaN.

Validation:

- Percentage symbols were removed.
- MISSING was replaced with NaN.
- Negative marks were converted to positive values.
- Marks above 100 were capped at 100.
- Both baseline_mark and latest_mark should now contain only values from 0 to 100 or missing values.

Tutoring Attendance

Rule:

- Values must be numeric percentages.
- Valid values must be between 0 and 100.
- Missing values may be retained.

Validation:

- Percentage symbols were removed.
- Negative values were converted to positive values.
- Values above 100 were capped at 100.
- The minimum and maximum values were checked after cleaning.

Videos Watched

Rule:

- Values must be numeric whole-number counts.
- Values cannot be negative.
- Missing values may be retained.

Validation:

- Negative values were converted to positive values.
- Missing values were identified.
- The column should contain no negative values.

Watch Time

Rule:

- Values must be numeric.
- Decimal separators must use periods.
- Values cannot be negative.
- Missing values may be retained.

Validation:

- Commas were replaced with periods.
- Values were converted to floating-point numbers.
- Negative values were converted to positive values.
- Descriptive statistics and null values were checked.
- The column should contain no negative or non-numeric values.

Capacity Fields

The following fields were validated:

- active_tutors
- weekly_tutor_hours
- waiting_list

Rule:

- Values must be numeric whole numbers.
- Values cannot be negative.
- Missing values may be retained.

Validation:

- Values were converted to nullable integer format.

- Duplicate or incomplete support-capacity rows were removed.
- The remaining records should contain valid numeric capacity values.

Cost Fields

Field: estimated_cost_per_extra_hour

Rule:

- Values must be numeric.
- Values cannot be negative.
- Missing values may be retained.
- Values should represent realistic monetary amounts.

Validation:

- The field was converted to nullable integer format.
- The remaining cost values should be checked to confirm that none are negative or missing unexpectedly.

Registration Conversion

Field: converted_to_registration

Rule:

Values must be either:

- Yes
- No

Validation:

- Y and 1 were converted to Yes.
- N and 0 were converted to No.
- The column should now contain only Yes, No, or missing values.